

Address Book (Python CLI)

A simple interactive command-line address book written in Python. It lets you add, search, delete, and list contacts from a looping main menu.

Features

- Add a new contact (name, type, phone number)
- Prevent duplicate phone numbers
- Search contacts by name (partial match) or by number
- Delete a contact by name or by number
- Show all saved contacts
- Input validation for menu choices

Requirements

- Python 3.8 or newer
- No external libraries

How to Run

1. Clone or download this repository.
2. Open a terminal in the project folder.
3. Run:

```
python address_book.py
```

Rename the file to `address_book.py` if your file has a different name.

Menu

1. Add new contact
2. Search by name
3. Search by number
4. Delete contact by name
5. Delete contact by number

6. Show all contacts
7. Exit

Example

```
Please enter your choice: 1
Enter name: ahmad
Enter type: friend
Enter number: 0791111111
Contact added.
```

Adding the same number again prints:

```
Number already exists, process.
```

How It Works

- Contacts are stored in memory as a list of lists: `[name, type, number]`.
- The main menu runs inside a `while True` loop until the user chooses Exit.
- Search loops over the list and compares the name or number.
- Delete finds the matching index and removes it with `pop()`.

Skills Demonstrated

- Python lists and nested lists
- Loops and conditional logic
- User input handling
- Basic CRUD operations (Create, Read, Delete)

Limitations and Future Improvements

- Data is lost when the program closes. Next step: save contacts to a file (JSON or CSV).
- Add an update/edit option.
- Validate phone number format.
- Rewrite using functions or a `Contact` class (OOP).

Author

